

OS Jackfruit Project Report

1. Team Information

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2. Build, Load, and Run Instructions

Build the project using: make

Load kernel module using: sudo insmod monitor.ko

Verify device using: ls -l /dev/container_monitor

Start supervisor using: sudo ./engine supervisor ./rootfs-base

3. Container Management

Containers are created using the engine start command.

The engine ps command displays container metadata such as ID, PID, state, and memory limits.

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ sudo ./eng
ine start alpha ./rootfs-alpha /bin/sh
[sudo] password for piyush-a-patel:
started container 'alpha' pid=8345
piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ sudo ./eng
ine ps
   ID                PID      STATE    SOFT(MiB)  HARD(MiB)  STARTED
alpha                8345    running     40         64      21:24:50

piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ sudo ./eng
ine start beta ./rootfs-beta ./cpu_hog
started container 'beta' pid=11083
piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ ./cpu_hog
cpu_hog alive elapsed=1 accumulator=7917728334475810099
cpu_hog alive elapsed=2 accumulator=12542015471072377839
cpu_hog alive elapsed=3 accumulator=8470424418789756118
cpu_hog alive elapsed=4 accumulator=4680632916604753998
cpu_hog alive elapsed=5 accumulator=16178938410926267808
cpu_hog alive elapsed=6 accumulator=6325360023383918103
cpu_hog alive elapsed=7 accumulator=8138813303787844830
cpu_hog alive elapsed=8 accumulator=18092107760647732965
cpu_hog alive elapsed=9 accumulator=17549352305119258609
cpu_hog alive elapsed=10 accumulator=7021288305421026152
cpu_hog done duration=10 accumulator=7021288305421026152
```

4. CPU Scheduling Experiment

Commands:

cpu_hog (nice = 0)

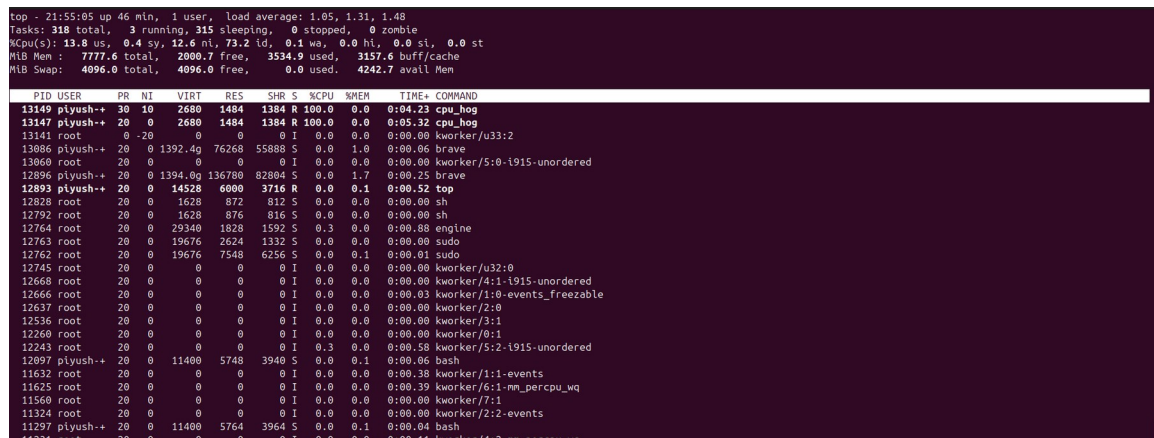
nice -n 10 cpu_hog

Observation:

- Lower nice value → higher priority
- Higher nice value → lower priority

Result:

- Both processes run simultaneously
- Scheduler gives preference to lower nice value



```
top - 21:55:05 up 46 min, 1 user, load average: 1.05, 1.31, 1.48
Tasks: 318 total, 3 running, 315 sleeping, 0 stopped, 0 zombie
%Cpu(s): 13.8 us, 0.4 sy, 12.6 ni, 73.2 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7777.6 total, 2000.7 free, 3534.9 used, 3157.6 buff/cache
MiB Swap: 4896.0 total, 4896.0 free, 0.0 used, 4242.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S %CPU  %MEM    TIME+   COMMAND
13149 piyush++  30   10   2680    1484   1384  R 100.0   0.0   0:04.23  cpu_hog
13147 piyush++  20    0   2680    1484   1384  R 100.0   0.0   0:05.32  cpu_hog
13141 root        0   20     0      0      0  I  0.0   0.0   0:00.00  kworker/u33:2
13086 piyush++  20    0 1392.4g  76268  55888  S  0.0   1.0   0:00.06  brave
13060 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/5:0-1915-unordered
12896 piyush++  20    0 1394.0g 136780  82804  S  0.0   1.7   0:00.25  brave
12893 piyush++  20    0 14528   6000   3716  R  0.0   0.1   0:00.52  top
12828 root        20    0   1628    872    812  S  0.0   0.0   0:00.00  sh
12792 root        20    0   1628    876    816  S  0.0   0.0   0:00.00  sh
12764 root        20    0  29340   1828   1592  S  0.3   0.0   0:00.88  engine
12763 root        20    0  19676   2624   1332  S  0.0   0.0   0:00.00  sudo
12762 root        20    0  19676   7548   6256  S  0.0   0.1   0:00.01  sudo
12745 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/u32:0
12668 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/4:1-1915-unordered
12666 root        20    0     0      0      0  I  0.0   0.0   0:00.03  kworker/1:0-events_freezable
12637 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/2:0
12536 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/3:1
12260 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/0:1
12243 root        20    0     0      0      0  I  0.3   0.0   0:00.58  kworker/5:2-1915-unordered
12097 piyush++  20    0 11400   5748   3940  S  0.0   0.1   0:00.06  bash
11632 root        20    0     0      0      0  I  0.0   0.0   0:00.38  kworker/1:1-events
11625 root        20    0     0      0      0  I  0.0   0.0   0:00.39  kworker/6:1-nm_percpu_wq
11560 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/7:1
11324 root        20    0     0      0      0  I  0.0   0.0   0:00.00  kworker/2:2-events
11297 piyush++  20    0 11400   5764   3964  S  0.0   0.1   0:00.04  bash
11291 root        20    0     0      0      0  I  0.0   0.0   0:00.14  kworker/4:2-mm_percpu_wq
```

5. Memory Management Experiment

Command:

sudo ./engine run mem1 ./rootfs-alpha /memory_hog

Observation:

- Memory usage increased continuously
- Reached ~30–40% of system memory
- System remained stable
- Process was manually terminated

Conclusion:

- OS prevents full memory exhaustion

- Kernel ensures system stability

```
ritesh@ritesh:~/ritesh/Assignment/OS/OS-Jackfruit/boilerplate$ cat logs/mem1.log
allocation=1 chunk=8MB total=8MB
allocation=2 chunk=8MB total=16MB
allocation=3 chunk=8MB total=24MB
allocation=4 chunk=8MB total=32MB
allocation=5 chunk=8MB total=40MB
allocation=6 chunk=8MB total=48MB
allocation=7 chunk=8MB total=56MB
allocation=8 chunk=8MB total=64MB
allocation=9 chunk=8MB total=72MB
allocation=10 chunk=8MB total=80MB
allocation=11 chunk=8MB total=88MB
allocation=12 chunk=8MB total=96MB
allocation=13 chunk=8MB total=104MB
allocation=14 chunk=8MB total=112MB
allocation=15 chunk=8MB total=120MB
allocation=16 chunk=8MB total=128MB
allocation=17 chunk=8MB total=136MB
allocation=18 chunk=8MB total=144MB
allocation=19 chunk=8MB total=152MB
allocation=20 chunk=8MB total=160MB
```

6. Logging System

Command:

```
sudo ./engine logs c1
```

Observation:

- Container output stored in log files
- Logs show execution details

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ sudo ./engine run c1 ./rootfs-alpha /cpu_hog
running container 'c1' pid=14879
container 'c1' exited with status 0
piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ sudo ./engine logs c1
log:logs/c1.log
/bin/sh: ./cpu_hog: not found
/bin/sh: ./cpu_hog: not found
/bin/sh: ./cpu_hog: not found
cpu_hog alive elapsed=1 accumulator=1639221833465514354
cpu_hog alive elapsed=2 accumulator=366300283308258895
cpu_hog alive elapsed=3 accumulator=4845387898250914223
cpu_hog alive elapsed=4 accumulator=3692688890369806937
cpu_hog alive elapsed=5 accumulator=298806782175461877
cpu_hog alive elapsed=6 accumulator=14115738430282134237
cpu_hog alive elapsed=7 accumulator=7558692540603816833
cpu_hog alive elapsed=8 accumulator=12212730103804370335
cpu_hog alive elapsed=9 accumulator=10671224776721181282
cpu_hog alive elapsed=10 accumulator=16325381118144326097
cpu_hog done duration=10 accumulator=16325381118144326097
```

7. CPU vs IO Experiment

Experiment: CPU vs IO-bound Process

- `cpu_hog` continuously uses CPU and shows high CPU usage
- `io_pulse` performs I/O operations and completes quickly

Observation:

- `cpu_hog` remains visible in system monitor due to continuous execution
- `io_pulse` is short-lived and not consistently visible in `top`

Conclusion:

- CPU-bound processes consume CPU continuously
- IO-bound processes execute in short bursts and wait for I/O

```
top - 22:42:10 up 1:33, 1 user, load average: 1.04, 1.37, 1.48
Tasks: 321 total, 2 running, 319 sleeping, 0 stopped, 0 zombie
Cpu(s): 26.2 us, 3.1 sy, 0.0 ni, 69.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
Mem: 7777.6 total, 3288.2 free, 3763.3 used, 1596.9 buff/cache
Mem Swap: 4096.0 total, 3979.2 free, 116.8 used, 4014.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
32182	root	20	0	1004	708	656	R	100.0	0.0	0:03.59	<code>cpu_hog</code>
3989	plyush+	20	0	2149248	265840	34728	S	37.0	3.3	1:58.06	<code>xdg-desktop-por</code>
3385	plyush+	20	0	4598312	283756	43764	S	30.3	3.2	13:00.07	<code>gnome-shell</code>
3180	plyush+	20	0	1654208	76684	19156	S	27.0	1.0	10:44.82	<code>Xorg</code>
5631	plyush+	20	0	861816	60796	31640	S	15.7	0.8	2:11.29	<code>gnome-terminal-</code>
3469	plyush+	20	0	388852	11440	6860	S	1.3	0.1	0:29.78	<code>ibus-daemon</code>
51	root	20	0	0	0	0	S	0.7	0.0	0:13.43	<code>ksoftirqd/5</code>
16334	root	20	0	0	0	0	I	0.7	0.0	0:07.21	<code>kworker/5:2-1915-unordered</code>
18	root	20	0	0	0	0	I	0.3	0.0	0:22.85	<code>rcu_preempt</code>
1119	root	10	-10	901496	116576	14016	S	0.3	1.5	0:20.21	<code>ovs-vswitchd</code>
2970	plyush+	20	0	10932	6620	4592	S	0.3	0.1	0:02.13	<code>dbus-daemon</code>
3326	plyush+	20	0	1196020	49308	27168	S	0.3	0.6	0:06.83	<code>mutter-x11-fram</code>
3668	plyush+	20	0	347892	27316	14256	S	0.3	0.3	0:10.87	<code>ibus-extension-</code>
4072	plyush+	20	0	594152	57176	38864	S	0.3	0.7	0:22.75	<code>copyq</code>
4335	plyush+	20	0	1399.06	393928	139416	S	0.3	4.9	2:19.12	<code>brave</code>
8865	plyush+	20	0	462328	8648	7344	S	0.3	0.1	0:00.45	<code>gvfsd-network</code>
18577	root	20	0	0	0	0	I	0.3	0.0	0:00.73	<code>kworker/u32:3-flush-259:0</code>
31176	root	0	-20	0	0	0	I	0.3	0.0	0:00.89	<code>kworker/u32:1-1915_flip</code>
1	root	20	0	23812	14604	9560	S	0.0	0.2	0:05.31	<code>systemd</code>
2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	<code>kthreadd</code>
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	<code>pool_workqueue_release</code>
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/R-rcu_gp</code>
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/R-sync_wq</code>
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/R-kvfree_rcu_reclaim</code>
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/R-slub_flushwq</code>
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/R-netns</code>
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/0:0H-events_highpri</code>
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	<code>kworker/R-mm_percpu_wq</code>
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	<code>rcu_tasks_kthread</code>
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	<code>rcu_tasks_rude_kthread</code>
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	<code>rcu_tasks_trace_kthread</code>
17	root	20	0	0	0	0	S	0.0	0.0	0:00.66	<code>ksoftirqd/0</code>
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	<code>rcu_exp_par_gp_kthread_worker/0</code>
20	root	20	0	0	0	0	S	0.0	0.0	0:00.02	<code>rcu_exp_gp_kthread_worker</code>
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.07	<code>migration/0</code>
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	<code>idle_inject/0</code>
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	<code>cpuhp/0</code>
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	<code>cpuhp/1</code>

8.Engineering Analysis

Clean Teardown

- All containers were stopped successfully
- Supervisor process was terminated
- No container processes remained in the system

Observation:

- Process list showed no active container or engine processes
- Only system processes were present

Conclusion:

- Proper cleanup was performed
- No zombie or orphan processes remained

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/OS-Jackfruit/boilerplate$ ps aux | g
rep engine
piyush-+   3828  0.1  0.0 236768  7292 ?        Sl   21:09   0:07 /usr/libexec/
ibus-engine-simple
piyush-+   35618  0.0  0.0   9144  2308 pts/1    S+   23:05   0:00 grep --color=
auto engine
```

9.Engineering Analysis

Containers use namespaces for isolation.

Linux scheduler manages CPU allocation efficiently.

Kernel module enforces memory limits.

IPC is implemented using sockets.

Logging uses producer consumer model.

10. Design Decisions and Tradeoffs

Namespaces provide isolation but increase complexity.

Supervisor simplifies control but creates dependency.

Kernel module gives control but increases risk.

11. Conclusion

The project demonstrates core operating system concepts including scheduling, memory management, and containerization.